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(54) COIL COMPONENT

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(57) **ABSTRACT**

In a coil component in which a terminal electrode including a conductor film including a sintered body of a conductive paste is provided on a mounting surface of a core facing a mounting board side, a fixing strength of the terminal electrode is increased. Mounting surfaces include raised portions having tops inside end edges of the mounting surfaces in a width direction orthogonal to the axial direction of the winding core, and the raised portions form inclined surfaces extending downward from the tops toward the end edges. When the conductive paste to be conductor films is applied to the mounting surfaces, since the raised portions are on the mounting surfaces, an applied thickness of the conductive paste can be increased, and the conductor films including a sintered body of the conductive paste can be formed thick, thus increasing the fixing strength of terminal electrodes to the mounting surfaces.

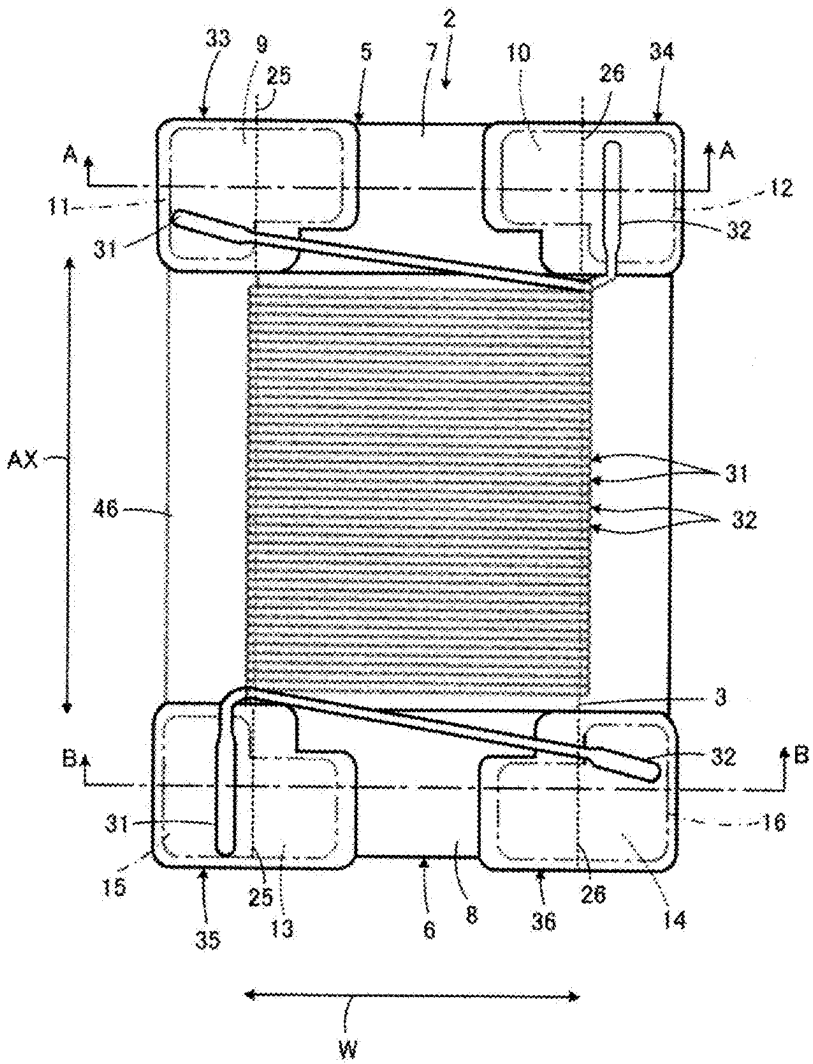

FIG. 1

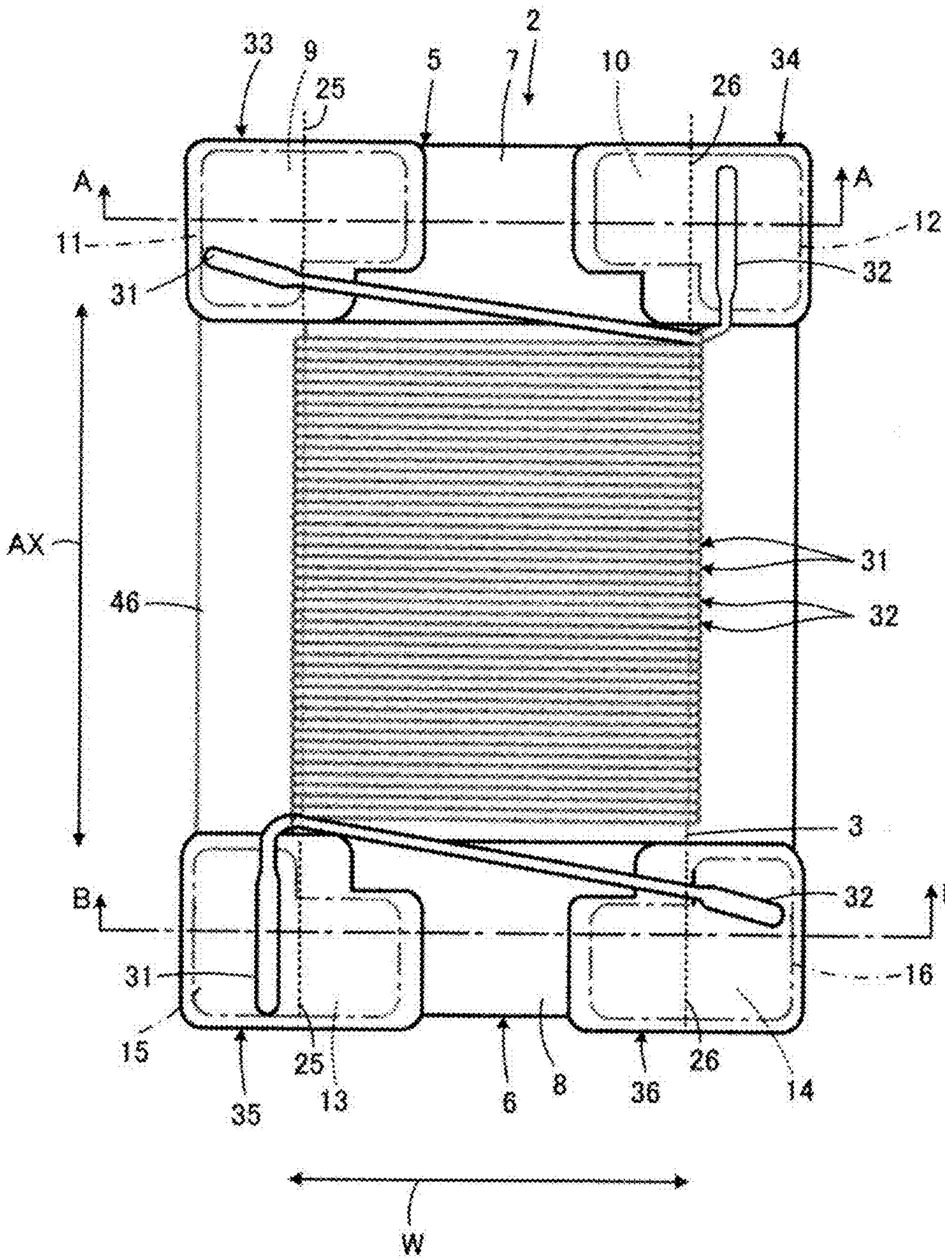
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FIG. 2A

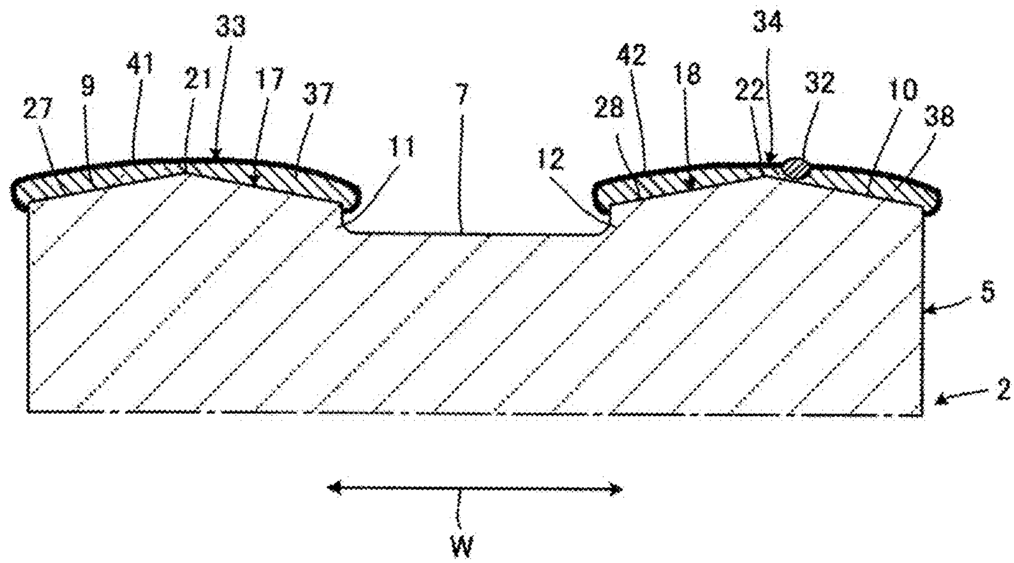
33
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FIG. 2B

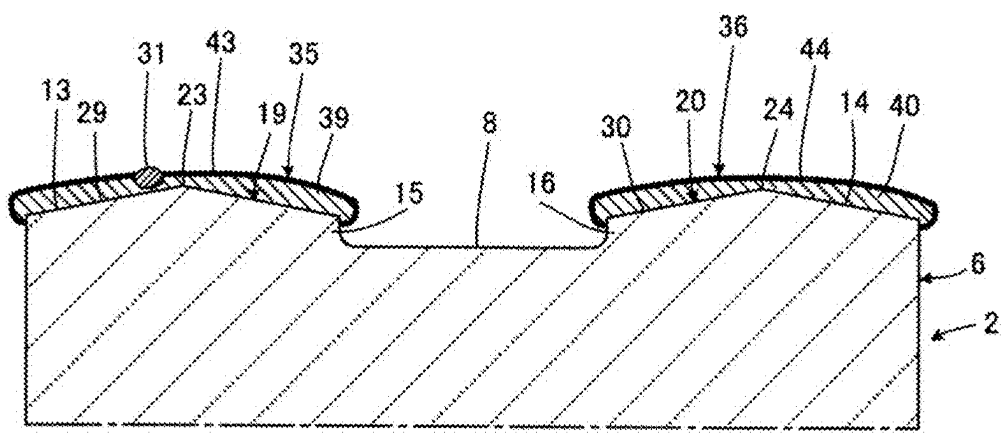
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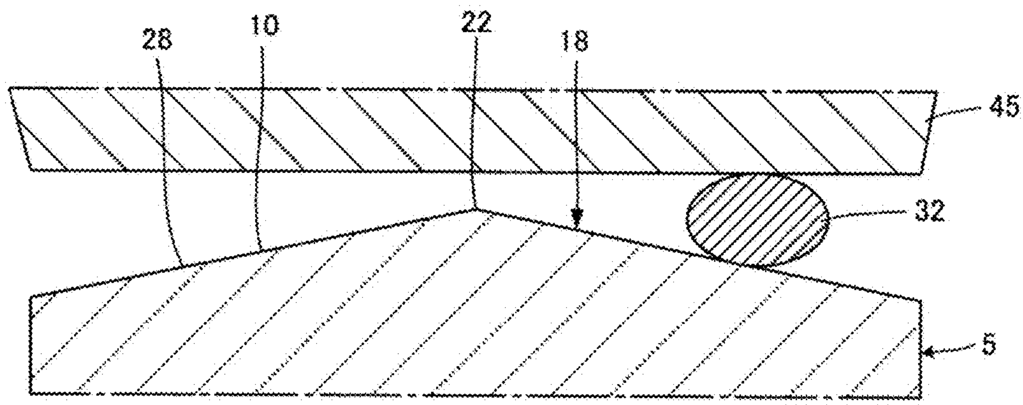
FIG. 3



COIL COMPONENT

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims benefit of priority to Japanese Patent Application No. 2024-033764, filed Mar. 6, 2024, the entire content of which is incorporated herein by reference.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a coil component including a core having a winding core around which a wire is wound and flanges provided at ends in an axial direction of the winding core, and a terminal electrode provided at the flange and to which the wire is connected, and particularly relates to a structure of a portion of the flange where the terminal electrode is provided.

Background Art

[0003] For example, Japanese Patent Application Laid-Open No. 2017-041589 discloses a coil component including a core having a winding core around which a wire is wound and flanges provided at ends of the winding core. A terminal electrode is provided on a mounting surface of the flange facing a mounting board side at the time of mounting. A wire is connected to the terminal electrode by thermocompression bonding. The terminal electrode includes a conductor film including a sintered body of a conductive paste applied to the mounting surface.

SUMMARY

[0004] The conductive paste to be the conductor film included in the terminal electrode is applied to the mounting surface by, for example, a dipping method. In the coil component described in Japanese Patent Application Laid-Open No. 2017-041589, as shown in the drawings of Japanese Patent Application Laid-Open No. 2017-041589, the mounting surface on which the terminal electrode is provided is flat. When the mounting surface is flat as described above, a thickness of the conductive paste applied thereto may not be sufficiently secured. In this case, there is a problem that a fixing strength of the terminal electrode to the mounting surface is lowered.

[0005] Therefore, the present disclosure provides a coil component capable of increasing the fixing strength of the terminal electrode to the mounting surface.

[0006] The present disclosure is directed to a coil component including a core having a winding core extending in an axial direction and flanges provided at ends in the axial direction of the winding core; a wire wound around the winding core; and a terminal electrode to which the wire is connected.

[0007] The flange has a mounting surface facing a mounting board side at the time of mounting. The terminal electrode includes a conductor film including a sintered body of a conductive paste applied to the mounting surface.

[0008] Also, the present disclosure is characterized in that the mounting surface is provided with a raised portion having a top located inside an end edge of the mounting surface in a width direction orthogonal to the axial direction, and the raised portion forms an inclined surface extending downward from the top toward the end edge.

[0009] According to the present disclosure, when the conductive paste to be the conductor film included in the terminal electrode is applied to the mounting surface by, for example, the dipping method, the raised portion is provided on the mounting surface, so that an applied thickness of the conductive paste can be increased. Therefore, the conductor film including the sintered body of the conductive paste can be formed thick, and the fixing strength to the mounting surface of the terminal electrode including the conductor film can be increased.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** is a bottom view showing an appearance of a coil component **1** according to an embodiment of the present disclosure from a mounting surface side;

[0011] FIG. **2**A is a sectional view taken along line A-A in FIG. **1**, and FIG. **2**B is a sectional view taken along line B-B in FIG. **1**; and

[0012] FIG. **3** is a sectional view for explaining a step of thermocompression bonding a wire **32** to a terminal electrode.

DETAILED DESCRIPTION

[0013] A coil component **1** according to an embodiment of the present disclosure will be described with reference to FIGS. **1**, **2**A, and **2**B.

[0014] The coil component **1** constitutes, for example, a common mode choke coil, and includes a core **2** including, for example, ferrite such as Ni—Zn-based ferrite, alumina, or resin containing metal magnetic powder. As shown in FIG. **1**, the core **2** includes a winding core **3** extending in an axial direction AX, and a first flange **5** and a second flange **6** respectively provided at first and second ends opposite to each other in the axial direction AX of the winding core **3**. The shown winding core **3** has, for example, a quadrangular cross-sectional shape, but may have a polygonal shape such as a hexagonal shape, a circular shape, an elliptical shape, or a shape obtained by combining these shapes.

[0015] Bottom surfaces **7** and **8** of the first flange **5** and the second flange **6** facing a mounting board (not shown) side at the time of mounting appear in FIG. **1**.

[0016] On the bottom surface **7** of the first flange **5**, two mounting surfaces **9** and **10** facing the mounting board side are arranged side by side in a width direction W orthogonal to the axial direction AX. As well shown in FIGS. **2**A and **2**B, the mounting surface **9** is located at a protrusion **11** raised from the bottom surface **7** with a step. The other mounting surface **10** is located at a protrusion **12** raised from the bottom surface **7** with a step.

[0017] Similarly, on the bottom surface **8** of the second flange **6**, two mounting surfaces **13** and **14** facing the mounting board side are arranged side by side in the width direction W. The mounting surface **13** is located at a protrusion **15** raised from the bottom surface **8** with a step. The other mounting surface **14** is located at a protrusion **16** raised from the bottom surface **8** with a step.

[0018] By respectively positioning the mounting surfaces **9**, **10**, **13**, and **14** on the protrusions **11**, **12**, **15**, and **16**, a step of applying a conductive paste described later can be easily performed by a dipping method.

[0019] As well shown in FIG. 2A, the mounting surface **9** is provided with a raised portion **17** that rises, and the mounting surface **10** is provided with a raised portion **18** that rises. Further, as well shown in FIG. 2B, the mounting surface **13** is provided with a raised portion **19** that rises, and the mounting surface **14** is provided with a raised portion **20** that rises. The raised portions **17**, **18**, **19**, and **20** respectively have tops **21**, **22**, **23**, and **24**.

[0020] As shown in FIG. **1**, the tops **21** and **23** are located on a virtual broken line **25**, and the tops **22** and **24** are located on a virtual broken line **26**. That is, the top **21** of the raised portion **17** is located inside an end edge of the mounting surface **9** in the width direction W, the top **22** of the raised portion **18** is located inside an end edge of the mounting surface **10** in the width direction W, the top **21** of the raised portion **19** is located inside an end edge of the mounting surface **13** in the width direction W, and the top **24** of the raised portion **20** is located inside an end edge of the mounting surface **14** in the width direction W.

[0021] The top **21** of the raised portion **17** has a portion extending linearly along the virtual broken line **25**, that is, a portion extending linearly in parallel with the axial direction AX of the winding core **3**. The top **22** of the raised portion **18** has a portion extending linearly along the virtual broken line **26**, that is, a portion extending linearly in parallel with the axial direction AX of the winding core **3**. The top **23** of the raised portion **19** has a portion extending linearly along the virtual broken line **25**, that is, a portion extending linearly in parallel with the axial direction AX of the winding core **3**. The top **24** of the raised portion **20** has a portion extending linearly along the virtual broken line **26**, that is, a portion extending linearly in parallel with the axial direction AX of the winding core **3**.

[0022] As shown in FIG. **2**A, the raised portion **17** forms an inclined surface **27** extending downward from the top **21** toward the end edge of the mounting surface **9**, and the raised portion **18** forms an inclined surface **28** extending downward from the top **22** toward the end edge of the mounting surface **10**. Similarly, as shown in FIG. **2**B, the raised portion **19** forms an inclined surface **29** extending downward from the top **23** toward the end edge of the mounting surface **13**, and the raised portion **20** forms an inclined surface **30** extending downward from the top **24** toward the end edge of the mounting surface **14**.

[0023] The inclined surface **27** of the raised portion **17** has a gradient viewed in a section (not shown) parallel to the axial direction AX of the winding core **3** larger than that viewed in a section (appears in FIG. **2**A) perpendicular to the axial direction AX of the winding core **3**. Similarly, the inclined surface **28** of the raised portion **18** has a gradient viewed in a section parallel to the axial direction AX of the winding core **3** larger than that viewed in a section perpendicular to the axial direction AX of the winding core **3**, the inclined surface **29** of the raised portion **19** has a gradient viewed in a section parallel to the axial direction AX of the winding core **3** larger than that viewed in a section perpendicular to the axial direction AX of the winding core **3**, and the inclined surface **30** of the raised portion **20** has a gradient viewed in a section parallel to the axial direction AX of the winding core **3** larger than that viewed in a section perpendicular to the axial direction AX of the winding core **3**.

[0024] In addition to the core **2** described in detail above, the coil component **1** includes a first wire **31** and a second wire **32** wound around the winding core **3**, a first terminal electrode **33** and a third terminal electrode **35** to which ends of the first wire **31** are respectively connected, and a second terminal electrode **34** and a fourth terminal electrode **36** to which ends of the second wire **32** are respectively connected. As shown in FIGS. **2**A and **2**B, the terminal electrodes **33** to **36** respectively include conductor films **37** to **40** including a sintered body of a conductive paste applied to each of the mounting surfaces **9**, **10**, **13**, and **14**.

[0025] In order to form the conductor films **37** to **40**, for example, a step of applying the conductive paste by the dipping method is performed. As the conductive paste, for example, a paste containing glass and silver is used. As described above, since the mounting surfaces **9**, **10**, **13**, and **14** are located at the protrusions **11**, **12**, **15**, and **16**, the conductive paste can be easily applied to the mounting surfaces **9**, **10**, **13**, and **14** by the dipping method.

[0026] The conductor films **37** to **40** obtained by firing the conductive paste applied as described above can be made relatively thick due to the presence of the raised portions **17** to **20** as shown in FIGS. **2**A and **2**B. Thus, fixing strengths of the terminal electrodes **33** to **36** to the mounting surfaces **9**, **10**, **13**, and **14** can be increased. In this embodiment, the conductor films **37** to **40** are thicker at portions located around the tops **21** to **24** of the raised portions **17** to **20** than at portions located on tops **21** to **24** of the raised portions **17** to **20**. This is particularly effective in increasing resistance of the terminal electrodes **33** to **36** to an external force in the width direction W.

[0027] Plating films **41** to **44** including, for example, a nickel plating layer and a tin plating layer on the nickel plating layer is formed on the conductor films **37** to **40** as necessary. In FIGS. **2**A and **2**B, the plating films **41** to **44** are schematically shown by thick lines.

[0028] FIG. **3** is a view for explaining a thermocompression bonding step for connecting the wires **31** and **32** to the terminal electrodes **33** to **36**. FIG. **3** shows the mounting surface **10** as a representative of the mounting surfaces **9**, **10**, **13**, and **14** on which the terminal electrodes **33** to **36** are provided, and shows the second wire **32**. Since the thermocompression bonding step is performed substantially similarly for any of the terminal electrodes **33** to **36**, the thermocompression bonding step for the terminal electrode **34** provided on the mounting surface **10** will be described with reference to FIG. **3**. Note that in FIG. **3**, the conductor film **38** provided on the mounting surface **10** and the terminal electrode **34** including the conductor film **38** are not shown.

[0029] In the thermocompression bonding step, a heater chip **45** for thermocompression bonding is operated toward the mounting surface **10** with the wire **32** sandwiched between the heater chip **45** and the mounting surface **10**. Thus, the wire **32** is thermocompression bonded to the terminal electrode **34** (not shown in FIG. **3**).

[0030] At this time, it is preferable that the wire **32** is disposed at a position other than the top **22** on the inclined surface **28** of the raised portion **18**, and is connected to the terminal electrode **34** at this position. This is because when the wire **32** is located to overlap the top **22**, stress concentrates on a portion of the wire **32** in contact with the top **22**, and the wire **32** may be cut at the portion where the stress concentrates.

[0031] The wire **32** is preferably connected to the terminal electrode **34** at a portion providing a height difference equal to or less than a diameter of the wire **32** before connection in the inclined surface **28**. With this configuration, a sufficient load can be applied from the heater chip **45** to the wire **32** at the time of pressure bonding, and pressure bonding quality can be stabilized. The diameter of the wire **32** is that "before connection" because the wire **32** after connection is flat, and the diameter of the wire **32** varies before and after connection.

[0032] Note that in an implemented product, for example, it is assumed that the diameter of the wire **32** before connection is 10 μm or more and 50 μm or less (i.e., from 10 μm to 50 μm), and the height difference of the inclined surface **28** is 1 μm or more and 10 μm or less (i.e., from 1 μm to 10 μm).

[0033] As illustrated in FIG. **1**, the coil component **1** may further include a top plate **46**. The top plate **46** includes, for example, substantially the same material as a material constituting the core **2**, and is stretched between surfaces respectively opposite to the bottom surfaces **7** and **8** of the first flange **5** and the second flange **6** to be fixed to the core **2**.

[0034] In this embodiment, for example, the inclined surface **28** extends from the top **22** of the raised portion **18** in at least two directions opposite to each other in the width direction W. The same applies to the inclined surfaces **27**, **29**, and **30**. When the inclined surfaces **27** to **30** are formed in this manner, a preferable pressure bonding portion of the wires **31** and **32** with respect to the terminal electrodes **33** to **36** can be more widely provided as compared with a case where the inclined surface extends only in one direction.

[0035] Although the present disclosure has been described with reference to the shown embodiment, various other modifications can be made within the scope of the present disclosure.

[0036] For example, the coil component according to the present disclosure may be a coil component constituting a transformer, a balun, or the like, or may be a coil component constituting a single coil, in addition to a coil component constituting the common mode choke coil as in the shown embodiment. Accordingly, the number of wires can also be changed depending on a function of the coil component, and the number of terminal electrodes provided on each flange can also be changed accordingly.

[0037] In the shown embodiment, the tops **21** to **24** of the raised portions **17** to **20** provided on the mounting surfaces **9**, **10**, **13**, and **14** have linearly extending portions, and the linearly extending portions extend parallel to the axial direction AX of the winding core **3**, however, the linearly extending portions may extend in a direction not parallel to the axial direction AX, for example, in a direction orthogonal to the axial direction AX.

[0038] The raised portion may not have a linearly extending portion, and may be provided to rise in a shape such as a conical shape or a truncated conical shape on the mounting surface.

[0039] Further, in the shown embodiment, on the mounting surfaces **9**, **10**, **13**, and **14**, the tops **21** to **24** of the raised portions **17** to **20** are at the highest position, but the highest portion may be present at a place other than the top.

[0040] In the shown embodiment, the raised portions **17** to **20** have a triangular section, and the tops **21** to **24** are located at tops of triangles, however, the top of the triangle providing the top may be chamfered, or a sectional shape of the entire raised portion may be a partial circle, a partial ellipse, or a shape similar thereto.

[0041] In configuring the coil component according to the present disclosure, partial replacement or combination of configurations can be made between different embodiments described in this specification.

[0042] The present disclosure includes the following embodiments.

[0043] <1> A coil component including a core having a winding core extending in an axial direction and flanges provided at ends in the axial direction of the winding core; a wire wound around the winding core; and a terminal electrode to which the wire is connected. The flange has a mounting surface facing a mounting board side at the time of mounting. The terminal electrode includes a conductor film provided on the mounting surface. Also, the mounting surface is provided with a raised portion having a top located inside an end edge of the mounting surface in a width direction orthogonal to the axial direction, and the raised portion has an inclined surface extending downward from the top toward the end edge.

[0044] <2> The coil component according to claim <1>, in which the mounting surface is located at a protrusion raised with a step from a bottom surface of the flange facing the mounting board side at the time of mounting.

[0045] <3> The coil component according to <1> or <2>, in which the conductor film has a portion located around the top of the raised portion thicker than a portion located on the top of the raised portion.

[0046] <4> The coil component according to any one of <1> to <3>, in which the top of the raised portion has a linearly extending portion.

[0047] <5> The coil component according to <4>, in which the linearly extending portion extends parallel to the axial direction of the winding core.

[0048] <6> The coil component according to <5>, in which the inclined surface extends from the top in at least two directions opposite to each other in the width direction.

[0049] <7> The coil component according to <4>, in which the linearly extending portion extends in a direction orthogonal to the axial direction of the winding core.

[0050] <8> The coil component according to any one of <1> to <7>, in which the inclined surface of the raised portion has a gradient viewed in a section parallel to the axial direction of the winding core larger than a gradient viewed in a section perpendicular to the axial direction of the winding core.

[0051] <9> The coil component according to any one of <1> to <8>, in which the wire is connected to the terminal electrode at a position other than the top in the inclined surface.

[0052] <10> The coil component according to any one of <1> to <9>, in which the wire is connected to the terminal electrode at a portion providing a height difference equal to or less than a diameter of the wire before connection in the inclined surface.

[0053] <11> The coil component according to <10>, in which the diameter of the wire before connection is 10 μm or more and 50 μm or less (i.e., from 10 μm to 50 μm), and the height difference of the inclined surface is 1 μm or more and 10 μm or less (i.e., from 1 μm to 10 μm).

What is claimed is:

1. A coil component comprising:

a core having a winding core extending in an axial direction and flanges at ends in the axial direction of the winding core;

a wire wound around the winding core; and
a terminal electrode to which the wire is connected, wherein
the flange has a mounting surface facing a mounting board side at the time of mounting,
the terminal electrode includes a conductor film on the mounting surface, and
the mounting surface includes a raised portion having a top inside an end edge of the mounting surface in a width direction orthogonal to the axial direction, and the raised portion has an inclined surface extending downward from the top toward the end edge.

2. The coil component according to claim **1**, wherein
the mounting surface is at a protrusion raised with a step from a bottom surface of the flange facing the mounting board side at the time of mounting.

3. The coil component according to claim **1**, wherein
a thickness of a portion of the conductor film around the top of the raised portion is thicker than a thickness of a portion of the conductor film on the top of the raised portion.

4. The coil component according to claim **1**, wherein
the top of the raised portion has a linearly extending portion.

5. The coil component according to claim **4**, wherein
the linearly extending portion extends in a direction parallel to the axial direction of the winding core.

6. The coil component according to claim **5**, wherein
the inclined surface extends from the top in at least two directions opposite to each other in the width direction.

7. The coil component according to claim **4**, wherein
the linearly extending portion extends in a direction orthogonal to the axial direction of the winding core.

8. The coil component according to claim **1**, wherein
a gradient of the inclined surface of the raised portion viewed in a cross section parallel to the axial direction of the winding core is larger than a gradient of the inclined surface of the raised portion viewed in a cross section perpendicular to the axial direction of the winding core.

9. The coil component according to claim **1**, wherein
the wire is connected to the terminal electrode at a position other than the top in the inclined surface.

10. The coil component according to claim **1**, wherein
the wire is connected to the terminal electrode at a portion of the inclined surface having a height difference equal to or less than a diameter of the wire before connecting the wire.

11. The coil component according to claim **10**, wherein
the diameter of the wire before connecting the wire is from 10 μm to 50, and
the height difference of the inclined surface is from 1 μm to 10 μm.

* * * * *